

Natural logarithmic functions



1 Differentiate the following functions:

a $y = \ln x$

c $y = \ln 6x$

e $y = \ln 4x - \ln 3$

g $y = x^4 + 6 \ln 7x$

i $y = 3 \ln x + 5 \ln 2x$

k $y = \ln (x^4 + 2)$

m $y = \ln (x^2 - 6x + 9)$

o $y = \ln \left((4x^3 + 8x^2 - 9)^3 \right)$

q $y = 4 \ln \left(\frac{x}{5} \right)$

s $y = \ln (8 - x) - 8$

u $y = \ln \left(\frac{8}{x+4} \right)$

b $y = 7 \ln x$

d $y = 2 \ln 3x$

f $y = 4 \ln 5x - x$

h $y = 4 \ln x - \frac{1}{x}$

j $y = \ln (7x + 3)$

l $y = \ln (x^2 + 7x + 5)$

n $y = \ln (2 + 4x + x^3)$

p $y = \ln \left(\frac{x}{7} \right)$

r $y = \ln \left(\frac{3}{x} \right)$

t $y = \ln (t^2 - 5t)$

v $y = \ln \left(\frac{4}{3x+4} \right)$

2 Consider the function $y = \ln \left(\frac{x-3}{x+3} \right)$.

a Let $u = \frac{x-3}{x+3}$. Rewrite y in terms of u .

c Find $\frac{dy}{du}$ in terms of x .

b Find $\frac{du}{dx}$.

d Find $\frac{dy}{dx}$.

3 Consider the function $y = \ln x^2$, where $x > 0$.

a Rewrite the function without powers.

c Find the value of x at which $y'(x) = \frac{1}{4}$.

b Hence determine $y'(x)$.

4 Consider the function $y = \ln (5x - 2)^4$.

a Rewrite the function without powers.

b Hence determine $y'(x)$.

c Find the exact value of x at which $y' = \frac{1}{3}$.

5 Differentiate the following functions:

a $y = \ln \sqrt{x}$

b $y = \ln \sqrt{7x}$

c $y = \ln \sqrt{2 - 5x}$

13 Suppose that $g(x) = \frac{\ln x}{f(x)}$, for some function  $f(x)$.

- a Find an expression for $g'(x)$ in terms of $f(x)$ and its derivative $f'(x)$.
- b If $f(e^3) = 4e^3$ and $f'(e^3) = 2$, find the value of $g'(x)$ when $x = e^3$.

14 Consider the functions $f(x) = k \ln x$ and $g(x) = \ln kx$, where $k > 1$ is a constant.

- a Find $f'(x)$.
- b Find $g'(x)$.
- c State how many times faster $f(x)$ is increasing compared to $g(x)$.

15 Given the following expressions:

- $\ln x = x - 1 - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots$
- $\frac{1}{x} = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - \dots$

Prove that $\frac{d}{dx}(\ln x) = \frac{1}{x}$.

Bases other than e

16 Consider the function $y = \log_3 x$.

- a Make x the subject.
- b Rewrite the expression for x with base e .
- c Find $\frac{dx}{dy}$.
- d Write $\frac{dx}{dy}$ in terms of x .
- e Hence find $\frac{dy}{dx}$.

17 Differentiate the following functions:

- a $f(x) = \log_2 x$
- b $f(x) = \log_2 4x$
- c $f(x) = \log_3 (2x - 4)$
- d $f(x) = \log_{10} (1 - x)$
- e $f(x) = \log_5 (4x + 2) - 6x$
- f $f(x) = \log_6 (2 - 7x)$